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⚙️ TECHNICAL WHITEPAPER

System Architecture & Tech Stack

Dokumen teknis lengkap tentang bagaimana Berstock dibangun,
di-host, di-deploy, dan kemungkinan self-hosting / white-label.

VERSI DOKUMEN

1.0 — Mei 2026

AUDIENCE

CTO · Tech Lead · IT Decision Maker

TIPE

Technical Whitepaper · Confidential

DISIAPKAN OLEH

Hendry Phang · Founder & Lead Developer

Dokumen ini menjelaskan secara teknis: (1) arsitektur sistem, (2) tech stack & vendor, (3) cara deployment, (4) data flow, (5) keamanan, (6) skalabilitas, dan (7) opsi white-label / self-hosting untuk enterprise client.

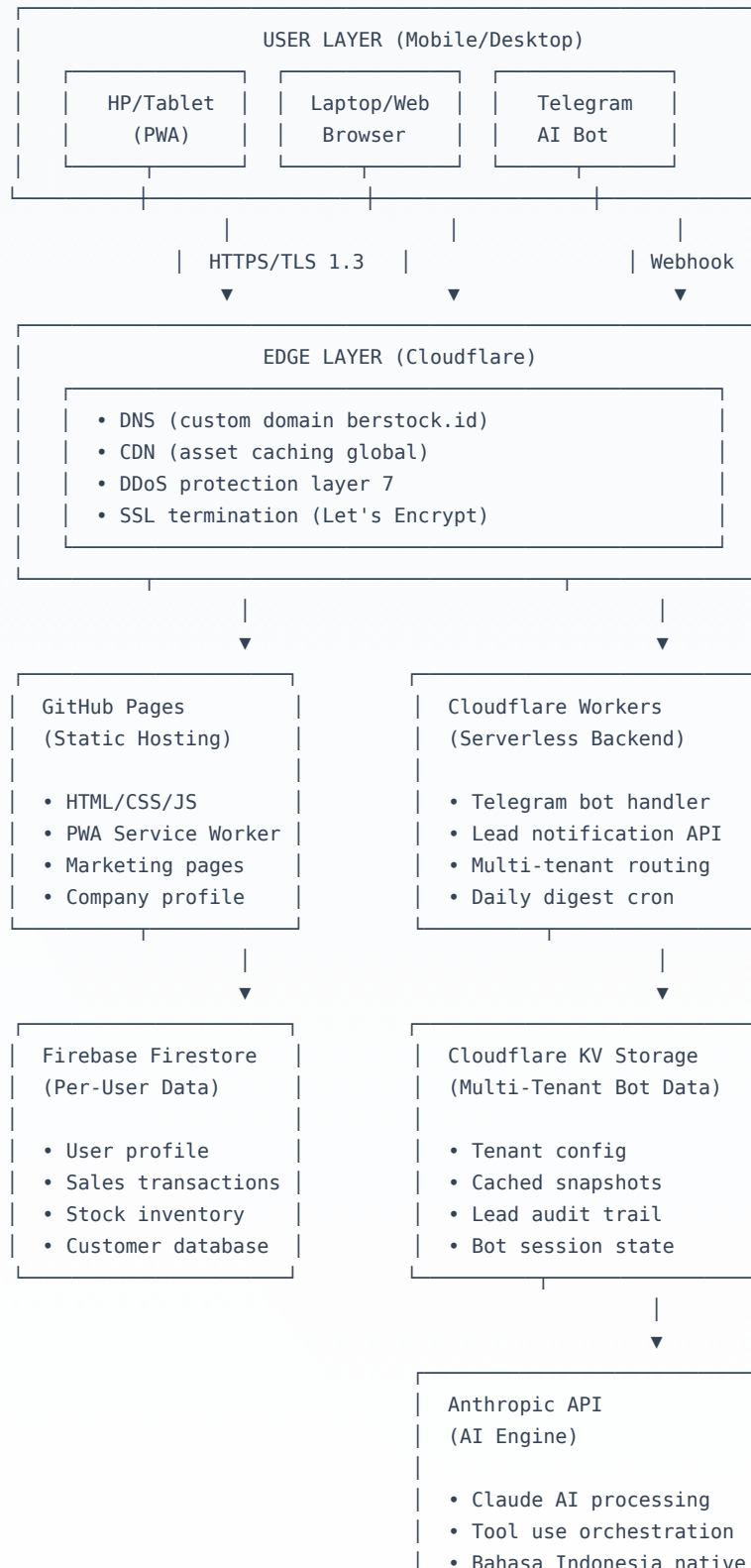


Arsitektur Sistem

High-level overview — bagaimana komponen Berstock terhubung



Diagram Arsitektur



Komponen Utama

Frontend

PWA — Vanilla HTML/CSS/JS
No framework, fast load

Backend Logic

Cloudflare Workers (V8 isolates)
Serverless edge runtime

Database

Firebase Firestore + Cloudflare KV
NoSQL document store

AI Engine

Anthropic Claude API
claude-sonnet-4-6 model

Hosting Static

GitHub Pages (FREE)
Branch deploy + CDN

CDN + DNS

Cloudflare
Edge caching global

Design Philosophy: Edge-first & Serverless

Tidak ada server EC2/VPS yang perlu di-maintain. Semua compute berjalan di edge runtime Cloudflare (auto-scale $0 \rightarrow \infty$) dengan low latency global.

Tech Stack & Vendor

Detail teknologi yang dipakai & alasan pemilihan

LAYER	TECHNOLOGY	VENDOR	REASONING PEMILIHAN
Frontend	HTML5 + Vanilla JS	—	No framework overhead. Bundle size minimal (~100KB gzipped). Fast load di HP entry-level.
UI Framework	Custom CSS	—	Brand-specific design. Tidak terikat library yang bisa break.
Charts	Chart.js v4.4.1	chartjs.org (OSS)	Industry standard, lightweight, accessible.
OCR	Tesseract.js v5	tesseract.org (OSS)	Untuk scan struk belanja di BerUang. Run di browser, tidak perlu server.
PWA	Service Worker	—	Install ke homescreen tanpa App Store. Offline-first via Cache API.
Auth	Firebase Auth v11	Google Cloud	Email/password + Google login. SOC 2 Type II certified. Free tier sangat generous.
Database (User Data)	Firebase Firestore	Google Cloud	NoSQL document store. Real-time sync built-in. Per-user isolation via Security Rules. AES-256 at-rest.
AI Backend	Cloudflare Workers	Cloudflare	V8 isolates (faster than container). Edge runtime di 200+ kota. Auto-scale tanpa konfigurasi.
AI Model	claude-sonnet-4-6	Anthropic	State-of-the-art reasoning + tool use. Bahasa Indonesia native. Constitutional AI (safer outputs).
Storage Bot	Cloudflare KV	Cloudflare	Eventually consistent key-value store. Multi-tenant via key prefix.
Bot Platform	Telegram Bot API	Telegram	Free, instant push, no approval delay (vs WhatsApp Business API).
DNS + CDN	Cloudflare DNS	Cloudflare	FREE tier unlimited. Anycast network 300+ locations. SSL gratis (Let's Encrypt + CF Universal SSL).
Hosting Static	GitHub Pages	GitHub (Microsoft)	FREE, unlimited bandwidth, custom domain support, auto-deploy from branch.
Domain Registrar	IDwebhost	IDwebhost (Indonesia)	Indonesian .id domain. ~Rp 240rb/tahun renewal.
Analytics	Google Analytics 4	Google	Industry standard. Free unlimited events. Conversion tracking + funnel.
Email	cPanel Email	IDwebhost	info@berstock.id native. Migrasi ke Google Workspace planned 2027.



Data Flow & Sequence

Bagaimana data mengalir antar komponen — 3 skenario utama



Skenario 1: User Transaksi di POS

```
[1] User klik "Bayar" di app PWA
      ↓
[2] saveSale() – write ke localStorage (instant, offline-safe)
      ↓
[3] Update state.products[].stok (decrement)
      ↓
[4] showStruk() – render thermal print preview
      ↓
[5] DEBOUNCED 30s: cloudSync.push() – POST ke Cloudflare Worker
      ↓
[6] Worker validates api_key + tenant_id
      ↓
[7] Worker writes ke Cloudflare KV (snapshot per tenant)
      ↓
[8] PARALLEL: Firebase Firestore sync (per-user document)
      ↓
[9] Auto-sync ✓ – data tersedia di multi-device + AI bot
```



Skenario 2: Owner Tanya AI Bot Telegram

```
[1] Owner kirim chat "Stok beras berapa?" ke @BerstockBot
      ↓
[2] Telegram → Webhook ke berstock-bot.workers.dev/webhook
      ↓
[3] Worker validasi X-Telegram-Bot-API-Secret-Token
      ↓
[4] Lookup tenant via chat_id mapping (KV)
      ↓
[5] Build context: snapshot terbaru dari KV + system prompt
      ↓
[6] Call Anthropic API (claude-sonnet-4-6) dengan tools=[12 tools]
      ↓
[7] Claude memilih tool: get_product_info({nama: "beras"})
      ↓
[8] Worker eksekusi tool – query KV snapshot
      ↓
[9] Hasil tool dikirim balik ke Claude
      ↓
[10] Claude format response Bahasa Indonesia natural
      ↓
[11] Worker → Telegram sendMessage API
      ↓
[12] Owner terima reply (~5 detik total)
```



Skenario 3: Lead Capture dari Web

```
[1] Visitor isi form lead di chat widget berstock.id
      ↓
[2] Submit → POST berstock-bot.workers.dev/api/lead
      ↓
[3] Worker validate input (nama, wa required)
      ↓
[4] PARALLEL EXECUTION:
├─ [4a] Send Telegram notif ke ADMIN_TELEGRAM_CHAT_ID
    |   (alert real-time ke owner Berstock)
├─ [4b] Save lead ke KV dengan TTL 90 hari
├─ [4c] GA4 trackEvent('lead_captured')
└─ [4d] User redirect ke wa.me dengan context pre-filled
      ↓
[5] Owner Berstock terima ping Telegram < 5 detik
      ↓
[6] Owner buka WA, chat lead langsung dengan context
```

⚡ Performance Karakteristik

- Frontend load: ~200ms (CDN cached)
- POS transaction save: instant (localStorage), 30s debounce ke cloud
- AI bot reply: 4-7 detik (tergantung kompleksitas tool use)
- Lead notif → Telegram: <5 detik



Keamanan & Privasi Data

Multi-layer security architecture sesuai best practice 2026

🔒 Encryption

LAYER	ENCRYPTION	PROVIDER
In Transit (User → Server)	TLS 1.3 (HTTPS)	Let's Encrypt + Cloudflare Universal SSL
At Rest (Database)	AES-256	Google Firestore
At Rest (KV)	AES-256	Cloudflare KV
API Keys (Workers)	Encrypted secrets	Cloudflare Workers Secrets
Telegram Webhook	HMAC-SHA256 secret token	Telegram Bot API

🛡️ Access Control

Per-User Isolation

Firestore Security Rules: user hanya bisa akses dokumen di path `users/{uid}/...` milik mereka.

Tenant Isolation (Bot)

Setiap tenant punya unique `api_key` + `tenant_id`. Cross-tenant query diblokir di Worker level.

Admin Whitelist

Hardcoded list di `auth.js`: 3 email allowed. CollectionGroup query untuk admin panel.

DDoS Protection

Cloudflare Layer 7 — auto-mitigate attack tanpa konfigurasi. Free tier sudah include.

📦 Data Privacy Commitment

🇮🇩 Data Sovereignty

- Data toko user disimpan di Firebase Firestore (region: asia-southeast1 — Singapore)
- Data bot (snapshots) di Cloudflare KV (region: distributed edge)
- Data **tidak dijual** ke pihak ketiga
- Data **tidak dipakai untuk training AI generic** (Anthropic enterprise terms)
- User export data dengan 1-klik (.json) — full data portability

📋 Backup & Recovery

DATA TYPE	BACKUP SCHEDULE	RETENTION
Firestore (user data)	Daily auto (Google native)	30 hari point-in-time recovery
KV (bot data)	Snapshot 6-jam	90 hari (TTL)
Source code	Git push real-time	Unlimited (GitHub)
Lead capture log	Real-time write ke KV	90 hari

🔑 Authentication Mechanisms

- **End User App**: Firebase Auth (email/password atau Google OAuth)
- **API Worker (sync)**: Bearer token `api_key` per tenant

- **Admin Panel:** Email whitelist + Firebase Auth
 - **Telegram Bot:** Webhook secret token validation
-



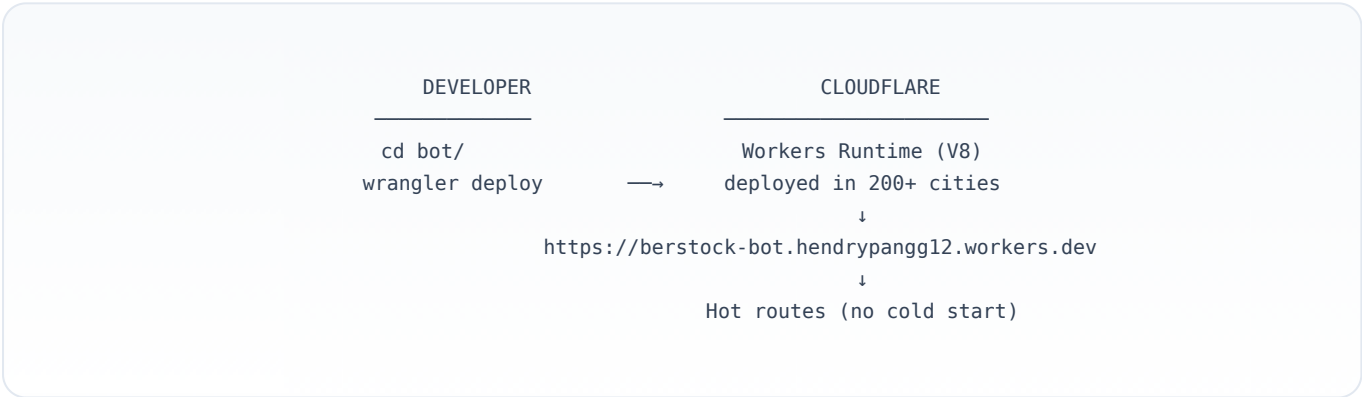
Deployment Pipeline

CI/CD flow dari git commit ke production

🚀 Frontend Deployment (GitHub Pages)



⚙️ Backend Deployment (Cloudflare Workers)



🔄 Cron Triggers

CRON SCHEDULE	TRIGGER TIME	ACTION
0 0 * * *	00:00 UTC (07:00 WIB)	Daily Digest — kirim ringkasan ke semua tenant Telegram

📊 Cost Structure (Monthly)

SERVICE	FREE TIER LIMIT	CURRENT USAGE	COST
Cloudflare Workers	100rb req/hari	<1k req/hari	Rp 0
Cloudflare KV	100rb read, 1k write/hari	<5k read/hari	Rp 0
Cloudflare DNS + CDN	Unlimited	—	Rp 0
Firebase Firestore	50k read, 20k write/hari	<1k both/hari	Rp 0
Firebase Auth	50k MAU	<100 user	Rp 0
GitHub Pages	1GB storage, 100GB bw	<100MB, low traffic	Rp 0
Anthropic API	Pay per use	~Rp 50-300/chat	Variable
Domain berstock.id	—	—	Rp 240rb/tahun

SERVICE	FREE TIER LIMIT	CURRENT USAGE	COST
Email (cPanel)	—	—	Rp 100rb/tahun
TOTAL FIXED COST			~Rp 28rb/bulan (scalable to 1000+ users with same cost)



Unit Economics

Karena 99% infrastruktur free-tier sampai 50K MAU, margin per klien (Rp 500rb/bulan) sangat sehat. Variable cost utama hanya Anthropic API yang ~5-10% dari subscription value.



White-Label & Self-Hosting

Bisakah orang lain pakai sistem ini sendiri? — penjelasan honest

? Pertanyaan: "Bisa di-share ke orang lain biar mereka customize sistem sendiri?"

🎯 Jawaban Singkat

Bisa, dengan 3 model: (1) White-Label, (2) Self-Hosted, (3) Source Code License. Masing-masing punya trade-off berbeda.

📦 Model 1: White-Label (Multi-Tenant Hosted)

ASPECT	DETAIL
Apa yang dapet klien	Sub-domain custom (misal: <code>klien1.berstock.id</code>) + branding sendiri
Yang di-host di	Infrastruktur Berstock (kami yang manage)
Customization yang bisa	Logo, warna, nama produk, system prompt AI bot, fitur flag on/off
Customization yang TIDAK bisa	Database schema, core business logic, integrasi pihak ketiga arbitrary
Pricing model	Setup fee + monthly/yearly + revenue share
Tech requirement klien	Tidak perlu — klien tinggal pakai

🏠 Model 2: Self-Hosted (Klien Manage Sendiri)

ASPECT	DETAIL
Apa yang dapet klien	Source code lengkap + dokumentasi + 1x setup assistance
Yang di-host di	Infrastruktur klien sendiri (Cloudflare account klien, Firebase project klien, dll)
Customization	FULL control — klien bisa ubah apa saja di kode
Cost klien (infra)	~Rp 0-100rb/bulan (sama free tier seperti kami)
Cost klien (Anthropic API)	Variable (klien bayar langsung ke Anthropic)
Tech requirement klien	Developer paham JS, Cloudflare Workers, Firebase. Bukan untuk non-technical.
Pricing model	One-time license fee + optional yearly support
Update / fitur baru	Klien pull dari repo Berstock (jika subscribe support) atau develop sendiri

📄 Model 3: Source Code License (Enterprise)

ASPECT	DETAIL
Apa yang dapet klien	Source code + perpetual license + opsional rebrand total
Yang di-host di	Infrastruktur klien (private cloud, on-prem, cloud provider pilihan)
Customization	UNLIMITED — klien bisa fork, rebrand, modifikasi total
Restriction	Tidak boleh re-distribute sebagai product baru kecuali ada perjanjian khusus
Pricing	Custom (mulai Rp 50jt+, tergantung scope)

ASPECT	DETAIL
Cocok untuk	Korporasi besar, government agency, SaaS B2B yang mau punya stack sendiri



Honest Recommendation

Untuk UMKM: Pilih hosted Berstock (Rp 500rb/bln) — tidak perlu manage infra.

Untuk software house: White-label (Model 1) — bisa rebrand untuk klien sendiri.

Untuk korporasi besar: Source code license (Model 3) — full control + private.

Untuk dev individu yang penasaran: Self-host (Model 2) — bayar setup, dapat code.



Skalabilitas & Limit

Berapa banyak user yang bisa di-handle? Apa bottleneck-nya?

Capacity Planning

TIER	ACTIVE USERS	INFRA COST	ACTION NEEDED
Stage 1: 0-100 user	Free tier semua	Rp 28rb/bulan	Nothing — current state
Stage 2: 100-1k user	Free tier masih cukup	Rp 28rb-100rb/bln	Optimize Anthropic prompt caching
Stage 3: 1k-10k user	Mungkin upgrade Firestore	Rp 500rb-2jt/bln	Migrate ke Firestore paid tier; tambah Anthropic budget
Stage 4: 10k-50k user	Cloudflare Workers Paid	Rp 5-15jt/bln	Workers Paid Plan (\$5/bln per 10jt req); Firestore Blaze plan
Stage 5: 50k+ user	Multi-region setup	Custom	Replicate Firestore multi-region; upgrade Cloudflare Enterprise

Performance Benchmarks

Frontend Load Time ~200ms (P50), ~500ms (P95) di 4G mobile Indonesia Lighthouse score 90+	POS Transaction Latency <50ms (localStorage write). Cloud sync 30s debounce. Offline-first design
AI Bot Reply Time 4-7 detik median (P50). 15s P95 untuk complex queries. Anthropic API latency dominates	Concurrent Users Theoretically unlimited (serverless). Tested: 1000 concurrent OK. Cloudflare Workers auto-scale

Bottlenecks & Mitigations

POTENTIAL BOTTLENECK	IMPACT	MITIGATION
Anthropic API rate limit	Bot reply gagal saat traffic spike	Prompt caching (95% hit), exponential backoff retry, fallback ke Haiku model
Firestore cost saat scale	Bill bisa naik kalau 100k+ user	Aggressive client-side caching, only sync deltas, paginate query
Cloudflare KV eventual consistency	Stale data ~60s window	Acceptable untuk use case kami (bukan financial transaction)
Telegram webhook timeout	Bot response >30s ditolak Telegram	<code>ctx.waitUntil()</code> async processing, ack dulu lalu kerja background
Single founder dependency	Bus factor = 1	Dokumentasi solid (CLAUDE.md), source code di GitHub, hire developer Q3 2026

Tertarik Diskusi Teknis Lebih Mendalam?

Dokumen ini overview teknis. Untuk diskusi spesifik (white-label, self-host, custom integration, security audit, atau enterprise license), Hendry siap call.



KONTAK TEKNIS

FOUNDER & LEAD DEV

Hendry Phang

WHATSAPP

+62 821-2484-8924

EMAIL BISNIS

info@berstock.id

WEBSITE

berstock.id



DISKUSI YANG BISA DIJADWALKAN

- White-label setup untuk software house bos
- Self-hosted deployment guide (untuk tim IT internal)
- Custom integration (API, ERP, payment gateway)
- Security audit / pentest readiness
- Enterprise license & private cloud deployment
- Multi-region failover untuk klien high-availability
- Custom AI tools development per industry vertical

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"Atur stok, untung pasti masuk."  

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